

Ubuntu

We install the **openssh-client** and **openssh-server** packages.

```
ubuntu@ubuntu-Standard-PC-Q35-ICH9-2009: ~ -- sudo apt install openssh-se

ubuntu@ubuntu-Standard-PC-Q35-ICH9-2009:~/.ssh$ cd
ubuntu@ubuntu-Standard-PC-Q35-ICH9-2009:~$ sudo apt install openssh-server openssh-client
openssh-client ya está en su versión más reciente (1:10.2p1-2ubuntu3.5).
fijado openssh-client como instalado manualmente.
Instalando:
  openssh-server

Instalando dependencias:
  ncurses-term openssh-sftp-server ssh-import-id

Paquetes sugeridos:
  molly-guard monkeysphere ssh-askpass

Resumen:
  Actualizando: 0, Instalando 4, Eliminando: 0, no actualizando: 11
  Tamaño de la descarga: 949 kB
  Espacio necesario: 7.585 kB / 8.997 MB disponible

¿Continuar? [S/n] s
```

We create the key using the following command: **ssh-keygen -t ed25519**

```
ubuntu@ubuntu-Standard-PC-Q35-ICH9-2009: ~

ubuntu@ubuntu-Standard-PC-Q35-ICH9-2009:~$ ssh-keygen -t ed25519
Generating public/private ed25519 key pair.
Enter file in which to save the key (/home/ubuntu/.ssh/id_ed25519):
Enter passphrase for "/home/ubuntu/.ssh/id_ed25519" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/ubuntu/.ssh/id_ed25519
Your public key has been saved in /home/ubuntu/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:fc8mHaefQj0ExYPmK/mpQ+WgKWGpftZhvDsz2A3ftNc ubuntu@ubuntu-Standard-PC-Q35-ICH9-2009
The key's randomart image is:
+--[ED25519 256]--+
|      o      |
|      o +    |
|   .  o o .   |
|  +   ..+ .   |
| + o = =Soo . |
| . o * O oo== |
|  o * B.==+.. |
|   . B =o+o..E |
|      oB   oo  |
+----[SHA256]-----+
ubuntu@ubuntu-Standard-PC-Q35-ICH9-2009:~$
```

Now, we need to add the public key of our Ubuntu client to the Debian server. In this way, authentication will be key-based. For this process, we need to temporarily enable password access on the Debian server, modifying the following line: PasswordAuthentication yes

```
47 PasswordAuthentication yes
1 KbdInteractiveAuthentication no
2 PubkeyAuthentication yes
3
4 PermitEmptyPasswords no
```

At this point, we use the following command on the Ubuntu client: `ssh-copy-id -p 2222 admin@www.debianserver.com` on the Ubuntu machine and provide the Admin user password. Once that command is executed, the public key of the Ubuntu client will be copied to the Debian server.

```
ubuntu@ubuntu-Standard-PC-Q35-ICH9-2009: ~
ubuntu@ubuntu-Standard-PC-Q35-ICH9-2009:~$ ssh-copy-id -p 2222 admin@www.debianserver.com
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: ssh-add -L
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
Authorized access only. All activity is monitored and logged
admin@www.debianserver.com's password:

Number of key(s) added: 1

Now try logging into the machine, with: "ssh -p 2222 'admin@www.debianserver.com'"
and check to make sure that only the key(s) you wanted were added.

ubuntu@ubuntu-Standard-PC-Q35-ICH9-2009:~$
```

Now, we must manually add the public key of the Ubuntu client on the Debian server. Specifically, in the file `/home/ftpuser/.ssh/authorized_keys`. In this way, the Ubuntu client is authorized to send files via SFTP with its public key.

```
root@debian-server:~# cat /home/ftpuser/.ssh/authorized_keys
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIJmgXS4LsjKvNxqZ20iU2R+jEVUVnJVIN6r05ADFV6xw ubuntu@ubuntu-Standard-PC-Q35-ICH9-2009
root@debian-server:~#
```

We modify the `/etc/ssh/sshd_config` file again. Specifically, the line: `PasswordAuthentication no`

```
47 PasswordAuthentication no
1 KbdInteractiveAuthentication no
2 PubkeyAuthentication yes
3
4 PermitEmptyPasswords no
```

Windows 10

Now, we need to add the public key of the Windows 10 Client to the Debian server so that SSH authentication is based on cryptographic keys and not on a local password. First, we create the cryptographic keys for the Windows 10 client.

```
Windows PowerShell
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Prueba la nueva tecnología PowerShell multiplataforma https://aka.ms/powershell

PS C:\Users\Win10cliente> ssh-keygen -t ed25519
Generating public/private ed25519 key pair.
Enter file in which to save the key (C:\Users\Win10cliente/.ssh/id_ed25519):
Created directory 'C:\Users\Win10cliente/.ssh'.
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in C:\Users\Win10cliente/.ssh/id_ed25519
Your public key has been saved in C:\Users\Win10cliente/.ssh/id_ed25519.pub
The key's fingerprint is:
SHA256:e1xRrnJ0TSpHJU6G757d/W4kpn73DU1GBwYEMI7roY win10cliente@DESKTOP-ITB0006
The key's randomart image is:
+--[ED25519 256]--+
|      .. oo.oo*.o |
|      .. .oB.= |
|      . ++* . |
|      o . *+ |
|      . S . +o .. |
|      . o + ..o |
|      . . o =.+* |
|      E o . =.o+B |
|      . .++ + |
+-----[SHA256]-----+
PS C:\Users\Win10cliente>
```

We temporarily allow the use of a local password to access the SSH server.

```
1 # To disable tunneled clear text passwords,
7 PasswordAuthentication yes
1 KbdInteractiveAuthentication no
2 PubkeyAuthentication yes
3
4 PermitEmptyPasswords no
```

Since the ssh-copy-id command does not exist in Windows, we execute the following command in Windows:

```
Get-Content $env:USERPROFILE\.ssh\id_ed25519.pub | ssh -p 2222  
admin@www.debianserver.com "mkdir -p ~/.ssh && chmod 700 ~/.ssh && cat >>  
~/.ssh/authorized_keys && chmod 600 ~/.ssh/authorized_keys"
```

Afterwards, we enter the passphrase for the public key.

```
Windows PowerShell
PS C:\Users\Win10cliente> Get-Content $env:USERPROFILE\.ssh\id_ed25519.pub | ssh -p 2222 admin@www.debianserver.com "m
s && chmod 600 ~/.ssh/authorized_keys"
Authorized access only. All activity is monitored and logged
Enter passphrase for key 'C:\Users\Win10cliente/.ssh/id_ed25519':
PS C:\Users\Win10cliente>
```

Now we manually copy the public key into /home/ftpuser/.ssh/authorized_keys from the Debian server. Therefore, the Windows 10 client will be ready to use the SFTP service through cryptographic keys.

```
root@debian-server:~# cat /home/ftpuser/.ssh/authorized_keys
ssh-ed25519 AAAAC3NzaC1lZD0IINTE5AAAAIJmgXS4LSjKvNxxqZ20iU2R+jEVUVnJVIN6r05ADFV6xw ubuntu@ubuntu-Standard-PC-Q35-ICH9-2009
ssh-ed25519 AAAAC3NzaC1lZD0IINTE5AAAAIAJG6asRrt+RUUJrCSZ2lrd3rD24Q/NiWYg76wpD4WZq win10cliente@DESKTOP-ITB0006
root@debian-server:~#
```

Now we can disable password access on the Debian server.

```
47 PasswordAuthentication no
   1 KbdInteractiveAuthentication no
   2 PubkeyAuthentication yes
```